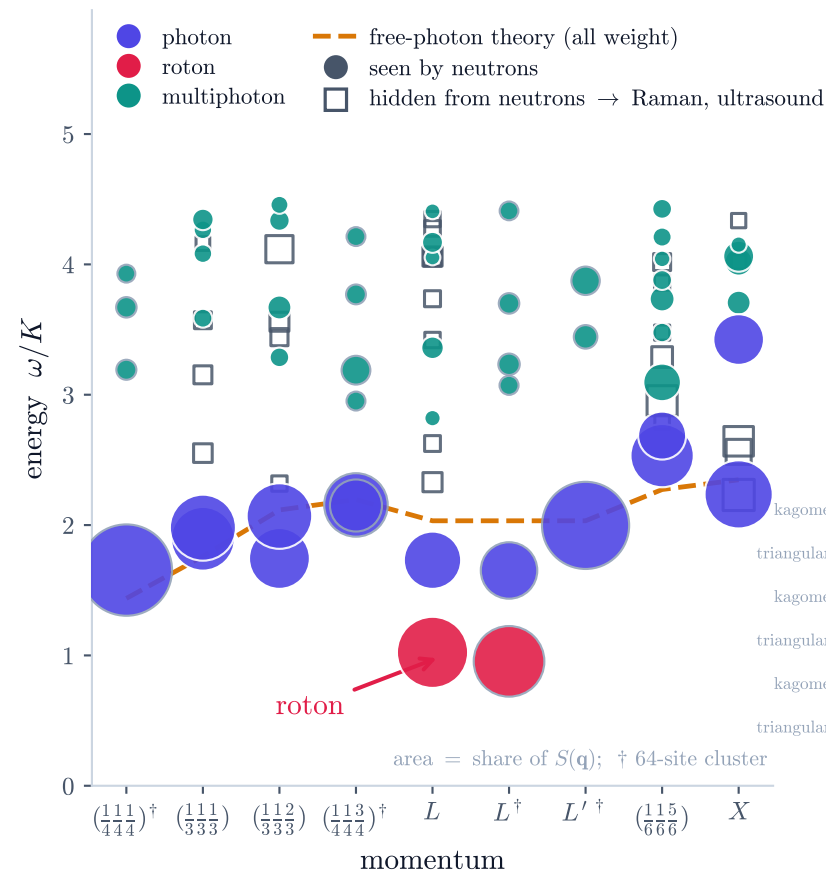
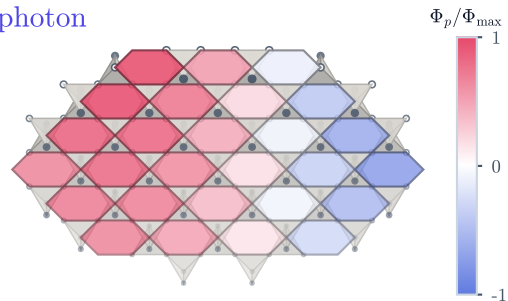
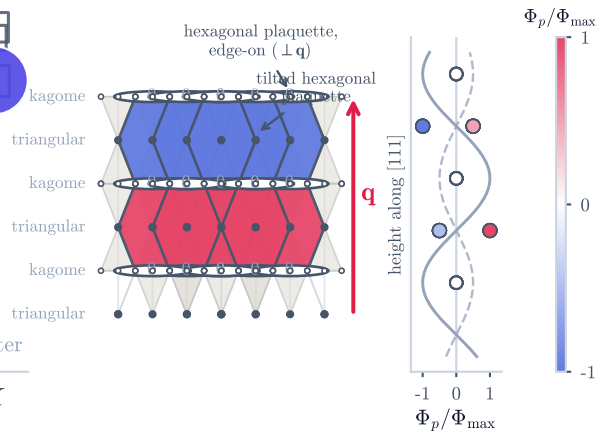




photon



roton



$\mathbf{q} \parallel [111]$: normal- $\parallel \mathbf{q}$ plaquettes: exactly zero flux;
 tilted families carry flux in ratio $(+1, -\frac{1}{2}, -\frac{1}{2})$ (sum 0),
 every plaquette reversing sign in the next region